

LAB MODULE #8: POWER CALIBRATION / HEAT BALANCE LABORATORY

LABORATORY OBJECTIVES:

1. Become familiar with the primary and secondary cooling system temperature monitoring instrumentation.
2. Increase reactor power to indicated levels and take temperature data.
3. Observe changes to the cooling system temperature parameters as it responds to the increasing heat load.
4. Plot **Core ΔT Power (kW)** -vs- **^{16}N Channel Power (kW)** at each power level.
5. Determine the **^{16}N Power Channel** calibration factor.

PRECONDITIONS:

1. Reactor Daily Startup Checklist completed and signed by Designated Senior Reactor Operator (DSRO).
2. Reactor startup has been authorized by DSRO.
3. Licensed Reactor Operator is on duty at the reactor control console.
4. Reactor power is stable at 250 kW.

EQUATIONS:

$$Q = \dot{m} \times C_p \times \Delta T$$

Where:

 Q = Power (BTU/HR) $\dot{m}$ = Coolant Mass Flow Rate (LBM/HR) C_p = Heat Capacity of Water (BTU/LBM/°F) ΔT = Change in temperature (°F) (e.g. across core or heat exchanger)*ASSUME: PRIMARY HEATUP RATE IS ZERO DUE TO MAINTENANCE OF CONSTANT TEMPERATURE FOR PRIMARY RETURN (T_6 & T_3).*

$$\Delta T_{core} = T_4 - T_2 \text{ (°F)}$$

See definitions of RTD's in #3 below.

$$\Delta T_{Heat-X} = T_5 - T_6 \text{ (°F)}$$

PROCEDURE:

1. Bring the reactor critical at power levels as indicated on **Table #1** referencing the ^{16}N Power Channel.
2. Allow temperatures to stabilize for 10 minutes prior to taking temperature data.
3. Record temperature values on **Table #1** as indicated on the following RTD's (Resistance Temperature Detectors):
 - T_2 – Pool Temperature (RTD is ~6 feet above top of fuel assemblies)
 - T_3 – Primary Cold (Return to pool)
 - T_4 – Primary Hot (Core outlet)
 - T_5 – Heat Exchanger (Hot) Primary Inlet
 - T_6 – Heat Exchanger (Cold) Primary Return
 - T_7 – Heat Exchanger (Cold) Secondary Inlet (from cooling tower condensate basin)
 - T_8 – Heat Exchanger (Hot) Secondary Outlet (to cooling tower spray header)
4. Measure Core ΔT and calculate reactor power (kW) using the mass flow equation above.

- Repeat steps #1 through #4 until a reactor power of 1000kW is reached.

Table #1 - Reactor Power and Cooling System Temperature Parameters:

Linear Channel (kW)	¹⁶ N Ch. Power (kW)	T ₂ (°F)	T ₃ (°F)	T ₄ (°F)	T ₅ (°F)	T ₆ (°F)	T ₇ (°F)	T ₈ (°F)	ΔT Core (T ₄ -T ₂)	ΔT Heat-X (T ₅ -T ₆)	True Power from Core ΔT
	250										
	500										
	750										
	1000										

ANALYSIS:

- Plot Reactor Power by **Core ΔT = T₄-T₂ (kW)** – vs- Power by ¹⁶N Channel (kW)
- Plot Reactor Power by **Heat Exchanger ΔT = T₅-T₆ (kW)** – vs- Power by ¹⁶N Channel (kW)
- Apply a linear fit to the Core ΔT data. The intercept should be set to zero. The value for the slope is the calibration factor for the ¹⁶N channel.
- Observe and explain the difference between the reactor power values obtained from the Core ΔT measurement relative to those for the Heat Exchanger ΔT measurement.

PULSTAR - Primary Cooling System Schematic

